

EVENTFLOW

EF-DOC-006

API & Application Service Design

Sprint 4 Consolidated Baseline - v1.0

Status: Approved Baseline

Field	Value
Document ID	EF-DOC-006
Version	1.0
Sprint	Sprint 4 - API & Application Service Design
Baseline dependency	Sprint 3 Database Baseline v1.0 / tag v0.4.0-database
Review state	Sprint 4 completeness & consistency review: PASS
Workflow validation	WF-001 through WF-014: 14/14 PASS
Promotion target	v1.0 - Approved Baseline

1. Purpose and Scope

This specification consolidates the accepted EventFlow Sprint 4 design decisions into a single implementation-grade API and application-service baseline. It defines how EventFlow clients, WordPress-authenticated administrators, public invitation guests, background workers, and provider webhooks interact with the accepted Sprint 3 domain and database model.

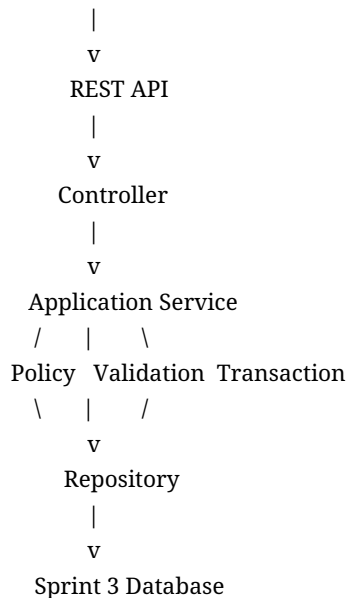
The document is normative for API contracts, service ownership, authorization boundaries, validation, concurrency, idempotency, transaction behavior, workflow orchestration, and error semantics. It does not replace the Sprint 3 database baseline and does not define production migration/rollback procedures; those remain a later operational documentation deliverable.

2. Architectural Principles

- REST controllers are transport adapters only; business rules live in application services and domain policies.
- Every Event-scoped operation carries explicit Event context and enforces cross-Event isolation before resource disclosure.

- Authentication, Event membership, capability authorization, and Event lifecycle authorization are separate checks.
- Application services own business invariants and transaction boundaries; repositories own persistence access.
- Historical state is preserved through cancellation, revocation, supersession, reversal, archival, and immutable snapshots rather than destructive rewriting.
- Optimistic concurrency, row locking, and idempotency are complementary mechanisms and are applied by operation type.
- External provider/network operations never execute inside long-lived database transactions.
- Asynchronous work preserves operation identity and is independently retry-safe.

Client / Admin UI / Guest UI / Reception / Worker



3. Accepted Sprint 4 Decisions

Decision	Status	Summary
API-001	Accepted	API conventions, versioning, JSON, pagination, filtering, sorting, request IDs, HTTP semantics.
API-002	Accepted	Authentication, principals, Event-scoped authorization, guest sessions, provider/webhook identity.
API-003A-E	Accepted	Application service catalogue and responsibilities.
API-004A-E	Accepted	REST endpoint catalogue across admin, guest, seating, communications, imports, audit and operations.
API-005A	Accepted	Canonical errors and layered validation.

Decision	Status	Summary
API-005B	Accepted	Concurrency, ETags, transactions, locking and idempotency.
API-006 / WF-001-WF-014	Accepted / Pass	Fourteen end-to-end workflow contracts validated.

4. API Conventions and Versioning

Primary namespace:

/wp-json/eventflow/v1/

Convention	Rule
Representation	application/json for normal request/response payloads; multipart/form-data for file uploads.
Field naming	snake_case.
Timestamps	ISO 8601 UTC on API boundaries; UI renders in Event timezone.
IDs	Administrative APIs may expose numeric IDs; public guest APIs do not require Event/Invitation IDs for normal operations.
Collections	items + pagination metadata; default page size 50, maximum 100.
Sorting	sort=field; descending with sort=-field; whitelist only.
Filtering	Explicit supported query parameters only; no arbitrary SQL-like expressions.
Search	q=... over safe indexed fields selected by repository/service.
Versioning	Backward-compatible additions remain v1; breaking contract changes require v2. Database schema version is independent.

5. Principal and Authorization Model

Principal type	Authentication source	Typical scope
wordpress_user	WordPress authenticated session / REST nonce	Global capability + Event membership + role/capability.
invitation_guest	Secure token bootstrap -> scoped guest session	Exactly one Event + Invitation; current token version.
background_job	System operation identity	Committed job/operation; preserves initiated_by_user_id.
provider_webhook	Provider-specific signature/secret verification	Verified provider callback only.
migration	Controlled system/CLI/deployment execution	Global operational scope.

Principal type	Authentication source	Typical scope
system	Internal EventFlow operation	Explicit system-owned actions only.

Authorization is evaluated as an intersection of identity, membership status/expiry, capability, resource Event scope, and current Event lifecycle. Cross-Event resources are generally concealed using `resource_not_found` rather than revealing existence.

6. Event Role Intent

Role	Intent
owner	Highest Event authority; configuration, memberships, archive/restore, communications, seating, reporting.
organizer	Most operational Event management excluding restricted ownership/security actions by default.
coordinator	Guest, seating and communications operations according to granted capabilities.
reception	Least-privilege attendee lookup, seating display, check-in; reversal only if explicitly elevated.
reporting	Read-only reports and permitted historical Event data.

7. Application Service Catalogue

Service	Primary responsibility
EventService	Event creation, lifecycle, activation, completion, archival, restoration; atomic initial configuration + owner membership.
VenueService	Reusable venue master data; no automatic historical snapshot rewrites.
EventConfigurationService	Event-specific behavior, branding, RSVP windows, seating mode, communication defaults.
MembershipService	Event membership, roles, suspension/revocation, primary ownership and continuity.
InvitationService	Invitation entitlement, capacity, contact, secure token lifecycle.
GuestAccessService	Token bootstrap, scoped guest session establishment, token-version revalidation.
RSVPService	Transactional whole-response RSVP reconciliation, capacity, attendee sync, guest edits.
AttendeeService	Organizer attendee administration, cancellation/restoration, primary-attendee commands.
SeatingGroupService	Invitation-derived and host-defined affinity groups; synchronization and membership.
SeatingService	Tables/seats, assignments, readiness, recommendations, manual moves, overrides.

Service	Primary responsibility
CheckInService	Immutable check-in actions, bulk check-in, effective state and reversals.
TemplateService	Draft/publish/version/archive templates, safe merge fields, preview rendering.
CampaignService	Campaign drafts, audience preview, schedule/queue, execution-time resolution and freeze.
MessageService	Logical Message state, provider attempts, retry eligibility and correlation.
ImportService	Upload/stage/map/normalize/validate/review/dry-run/apply/reconcile.
MigrationService	Migration readiness, checksums, locking, validation and recovery coordination.
AuditService	Single authoritative business/security audit writer; redaction of sensitive fields.

8. REST Endpoint Catalogue

Method	Endpoint	Purpose	Owner
GET	/events	List accessible Events	EventService
POST	/events	Create Event	EventService
GET/PATCH	/events/{event_id}	Read/update Event	EventService
POST	/events/{event_id}/ activate complete cancel archive restore	Lifecycle command	EventService
GET/POST	/venues	List/create Venues	VenueService
GET/PATCH	/venues/{venue_id}	Read/update Venue	VenueService
GET/PATCH	/events/{event_id}/ configuration	Event configuration	EventConfigurationService
GET/POST	/events/{event_id}/ memberships	List/grant memberships	MembershipService
PATCH	/events/{event_id}/ memberships/ {membership_id}	Change role/attributes	MembershipService
POST	/events/{event_id}/ memberships/ {membership_id}/ suspend reactivate revoke make-primary-owner	Membership commands	MembershipService
GET/POST	/events/{event_id}/ invitations	List/create Invitations	InvitationService
GET/PATCH	/events/{event_id}/ invitations/ {invitation_id}	Read/update Invitation	InvitationService

Method	Endpoint	Purpose	Owner
POST	/events/{event_id}/ invitations/ {invitation_id}/ activate revoke rotate- token archive restore	Invitation lifecycle/security	InvitationService
GET/POST	/events/{event_id}/ attendees	List/create Attendees	AttendeeService
GET/PATCH	/events/{event_id}/ attendees/{attendee_id}	Read/update Attendee	AttendeeService
POST	/events/{event_id}/ attendees/ {attendee_id}/cancel restore make-primary	Attendee commands	AttendeeService
POST	/public/invitations/ bootstrap	Invitation-token bootstrap	GuestAccessService
GET	/public/invitation	Guest-safe Event/Invitation context	GuestAccessService
GET/PUT	/public/invitation/ response	Read/submit complete RSVP	RSVPService
POST	/public/session/logout	End guest session	GuestAccessService
GET/POST	/events/{event_id}/ seating-groups	List/create seating groups	SeatingGroupService
GET/PATCH	/events/{event_id}/ seating-groups/ {group_id}	Read/update seating group	SeatingGroupService
GET/POST	/events/{event_id}/ tables	List/create Tables	SeatingService
GET/PATCH	/events/{event_id}/ tables/{table_id}	Read/update Table	SeatingService
GET/POST	/events/{event_id}/ tables/{table_id}/seats	Seats	SeatingService
GET	/events/{event_id}/ seating/readiness	Non-mutating seating preflight	SeatingService
POST	/events/{event_id}/ seating/ recommendations	Generate recommendations	SeatingService
GET	/events/{event_id}/ seating/ recommendations/ {recommendation_id}	Review recommendation	SeatingService
POST	/events/{event_id}/ seating/ recommendations/ {recommendation_id}/ apply	Apply approved recommendation	SeatingService

Method	Endpoint	Purpose	Owner
POST	/events/{event_id}/attendees/{attendee_id}/seating/move	Manual attendee move	SeatingService
POST	/events/{event_id}/seating-groups/{group_id}/move	Group move	SeatingService
GET	/events/{event_id}/reception/attendees	Least-privilege reception search	CheckInService/query
POST	/events/{event_id}/check-ins	Individual check-in	CheckInService
POST	/events/{event_id}/check-ins/bulk	Atomic bulk check-in	CheckInService
POST	/events/{event_id}/check-ins/{checkin_id}/reverse	Check-in reversal	CheckInService
GET/POST	/events/{event_id}/communication-templates	List/create Templates	TemplateService
GET/PATCH	/events/{event_id}/communication-templates/{template_id}	Read/update draft Template	TemplateService
POST	/events/{event_id}/communication-templates/{template_id}/publish new-version archive preview	Template commands	TemplateService
GET/POST	/events/{event_id}/campaigns	List/create Campaigns	CampaignService
GET/PATCH	/events/{event_id}/campaigns/{campaign_id}	Read/update Campaign draft	CampaignService
POST	/events/{event_id}/campaigns/{campaign_id}/audience-preview schedule queue cancel	Campaign commands	CampaignService
GET	/events/{event_id}/messages	List Messages	MessageService
GET	/events/{event_id}/messages/{message_id}	Message detail	MessageService
POST	/events/{event_id}/messages/{message_id}/retry	Retry logical Message	MessageService
POST	/webhooks/{provider}	Provider callback ingress	ProviderEventProcessor

Method	Endpoint	Purpose	Owner
GET/POST	/events/{event_id}/imports	List/create Import Jobs	ImportService
PUT	/events/{event_id}/imports/{import_job_id}/mapping	Column mapping	ImportService
POST	/events/{event_id}/imports/{import_job_id}/parse validate dry-run apply cancel	Import commands	ImportService
GET	/events/{event_id}/imports/{import_job_id}/rows result	Review/results	ImportService
GET	/events/{event_id}/audit	Read Event audit history	Audit query
GET	/system/health readiness	Operational health/readiness	System
GET	/system/migrations system/migrations/readiness	Migration status/readiness	MigrationService

9. Error and Validation Contract

```
{
  "error": {
    "code": "invitation_capacity_exceeded",
    "message": "This invitation allows up to 4 attendees.",
    "request_id": "ef_...",
    "details": {"capacity": 4, "submitted_count": 5}
  }
}
```

HTTP	Use
400	Malformed/uninterpretable request.
401	Authentication missing/invalid; invalid guest session/webhook auth.
403	Known principal lacks permission or current scoped permission.
404	Resource absent or deliberately concealed across Event boundary.
409	Valid request conflicts with current state.
412	If-Match/ETag stale.
422	Field/domain validation failed.
428	Required precondition omitted.
429	Rate limited; Retry-After when meaningful.
500	Unexpected internal failure with safe response.

HTTP	Use
503	Known temporary service unavailability.

10. Concurrency, Transactions and Idempotency

Operation class	Optimistic concurrency	Transaction / lock	Idempotency
Mutable admin resource	ETag + If-Match	Short transaction	Usually no
RSVP	Response revision	Invitation lock + atomic reconciliation	Required
Seating move	Expected assignment revision	Assignment + table/seat locks	Recommended for command
Check-in	Current-state validation	Atomic action write	Required
Campaign queue	Campaign state check	Freeze + Message creation transaction	Required
Message retry	Current state	Short attempt transaction	Required
Import apply	Job state	Batch transactions	Required
Provider webhook	N/A	Short dedupe/state transaction	PDM-017A dedupe

Idempotency keys are scoped by principal, operation/endpoint, and Event where applicable. Same key + same canonical request replays the original outcome; same key + materially different request returns idempotency_key_conflict. External provider calls occur after authoritative DB transactions commit.

11. Workflow Contract Validation

Workflow	Name	Result	Key contract
WF-001	Event Creation & Activation	PASS	Atomic Event + default configuration + primary owner membership; activation preconditions and venue snapshot.
WF-002	Invitation Creation & Secure Access	PASS	Return-once raw token; token-version-bound guest sessions; revocation/rotation invalidates access.
WF-003	Guest RSVP & Later Amendment	PASS	Whole-response reconciliation, scoped attendee refs, response revision, capacity lock, non-destructive cancellation.

Workflow	Name	Result	Key contract
WF-004	Organizer Attendee Administration	PASS	Capacity-safe organizer changes; explicit primary transfer; no seating/check-in mutation via attendee edit.
WF-005	Grouping & Seating Preparation	PASS	Invitation groups, affinity groups, flexible tables/seats, readiness preflight.
WF-006	Automatic Seating Recommendation	PASS	Explainable advisory recommendations, stale-plan fingerprint, manual protection, explicit host apply.
WF-007	Manual Seating Move & Override	PASS	Destination-first validation, history preservation, required-group override reason, stale assignment detection.
WF-008	Campaign Preparation & Dispatch	PASS	Execution-time audience resolution, fixed published template version, transactional freeze/message snapshots.
WF-009	Provider Callback & Retry	PASS	Provider-capability-aware retry, immutable attempts, dedupe, out-of-order transition validation.
WF-010	Reception Search & Check-in	PASS	Least-privilege reception DTO, attendee-level immutable check-in, atomic bulk, idempotent retry.
WF-011	Check-in Reversal	PASS	Additive reversal, one reversal per check-in, mandatory reason, elevated permission.
WF-012	Guest-list Import & Reconciliation	PASS	Stage/validate/dry-run; batched restart-safe domain-service application; final reconciliation.

Workflow	Name	Result	Key contract
WF-013	Membership & Primary Owner Transfer	PASS	Privilege escalation controls, owner continuity, non-expiring primary owner, stale-owner conflict check.
WF-014	Completion & Archival	PASS	Distinct completed/archived states, readiness checks, read-mostly archive, late historical evidence allowed.

12. Consolidated Sprint 4 Refinements

Ref	Normative refinement
S4-R01	Raw invitation token returned only at creation/rotation.
S4-R02	Guest sessions bound to current Invitation token version.
S4-R03	Public attendees use server-issued scoped attendee references.
S4-R04	RSVP exposes revision/stale-response conflict protection.
S4-R05	Primary-attendee change uses explicit transactional command.
S4-R06	Non-mutating seating readiness/preflight endpoint.
S4-R07	Seating recommendations carry input revision/fingerprint.
S4-R08	Required-group manual override requires a reason.
S4-R09	Seating move supports expected-current assignment state.
S4-R10	Campaign audience resolves at execution time.
S4-R11	Selected published Template version remains stable.
S4-R12	Provider adapters declare retry/idempotency/reconciliation capabilities.
S4-R13	Provider Event dedupe enforced transactionally/uniquely.
S4-R14	Reception uses dedicated least-privilege DTO.
S4-R15	Check-in reversal requires elevated permission and mandatory reason.
S4-R16	Import apply is restartable and batch-oriented.
S4-R17	Bulk invitation credential issuance cannot persist raw tokens.
S4-R18	Primary owner cannot automatically expire.

Ref	Normative refinement
S4-R19	Ownership transfer checks expected current primary owner.
S4-R20	Completion and archival readiness operations required.
S4-R21	Archived restore returns to completed; reactivation is separate.
S4-R22	Archive and permanent purge are distinct operations.

13. Event Lifecycle Contract

draft -> active -> completed -> archived

\ \n \-> cancelled (per permitted transition matrix)

Completed means ordinary operations are closed but narrowly controlled corrections or approved post-event communications may remain possible. Archived means historical/read-mostly; new business intent is blocked, while late evidence about previously committed operations (for example provider callbacks) may still be recorded. Restore is archived -> completed. Reopening to active is a separate privileged action with full readiness validation.

14. Public Guest Contract

- Raw invitation token is a bootstrap bearer credential; it is not repeatedly transmitted after session establishment.
- Guest Principal is bound to Event, Invitation and token version.
- Guest APIs do not allow the client to choose Event/Invitation scope.
- RSVP submission represents the complete guest-managed attendee set and is reconciled transactionally.
- Existing attendees use server-issued scoped references; new attendees use temporary client_ref values.
- Guest removal cancels/retires attendee participation rather than deleting history; event-day locked records require organizer intervention.
- Bootstrap is rate-limited and invalid/revoked/expired token responses avoid leaking credential state.

15. Seating Contract

- Invitation-derived groups are synchronized automatically; host affinity groups are independent and may overlap.
- Required, preferred and informational constraints have distinct behavior.
- Automatic seating is advisory and explainable; host approval is mandatory before authoritative changes.
- Normal recommendation mode protects manual assignments and explicit overrides.
- Apply rejects stale recommendation snapshots/fingerprints.

- Manual moves validate destination before superseding old assignment and preserve assignment history.
- Required-group violations require explicit override and reason.
- Seating remains independent of check-in; checked-in attendees may still be moved where Event policy allows.

16. Communications Contract

- Published template versions are immutable; edits create a new version.
- Campaign audience preview is dynamic and non-authoritative.
- Scheduled Campaigns keep their selected published Template version but resolve audience at execution time.
- Campaign queue transaction freezes definition, renders recipient-specific immutable Message snapshots, then commits before provider dispatch.
- Delivery retries create new Delivery Attempts for the same logical Message.
- Provider callbacks are verified, deduplicated with PDM-017A, normalized, and applied through a validated monotonic state-transition model.
- Campaign completion means processing completed, not universal successful delivery.

17. Import Contract

- Upload/staging never mutates production domain records.
- Mapping uses whitelisted application-level import fields, not arbitrary database columns.
- Dry-run and readiness are required before Apply.
- Apply is asynchronous, idempotent, batched and restartable.
- Production changes reuse domain-service invariants rather than direct bypass writes.
- Per-row outcomes are persisted for reconciliation and restart safety.
- Raw staging data is retention-controlled and may be purged after reconciliation.
- Bulk guest credential generation preserves return-once token semantics and must not persist raw tokens.

18. Audit and Operational Boundaries

AuditService is the authoritative business/security audit writer. Client-facing audit endpoints are read-only. Application/debug logs remain separate from business audit history. Production migration execution is not exposed as an ordinary v1 REST operation; controlled CLI/plugin-upgrade/deployment paths are preferred. System health/readiness endpoints expose only safe diagnostic information according to privilege.

19. Deferred / Future Concerns

Concern	Disposition
Full migration, upgrade, rollback and recovery runbook	Future operational document; mandatory before production migrations.
Permanent purge / retention durations	Separate governance/privacy/operations design; archive is not purge.
Durable seating recommendation/run persistence	Deferred until implementation proves a permanent run-history entity is required.
Formal negative seating constraints / keep-apart relationships	Deferred; host manual placement handles v1 unless product evidence justifies model extension.
API-level idempotency persistence table/cache	Implementation requirement; choose store before coding retry-sensitive endpoints.
Numeric response revision column	Optional schema optimization; current contract may derive revision/ETag until implementation needs dedicated column.
Reactivation endpoint	Separate privileged lifecycle operation if product requires reopening completed Events.
Partial bulk check-in mode	Deferred; v1 default is atomic all-or-nothing.

20. Sprint 4 Consolidation Verdict

The consolidated API and application-service design is internally consistent with the approved Sprint 3 database baseline. All fourteen workflow contracts passed. No blocking architectural contradiction remains. S4-R01 through S4-R22 are incorporated as normative refinements in this document.

Review gate	Result
Architecture	PASS
Service boundaries	PASS
API consistency	PASS
Authorization and Event isolation	PASS
Error/validation contract	PASS
Concurrency and transactions	PASS
Idempotency	PASS
Historical integrity	PASS
Workflow coverage	14/14 PASS
Sprint 3 compatibility	PASS
Blocking issues	NONE
Document state	v1.0 Approved Baseline

Promotion recommendation: after final document-level QA and user approval, promote EF-DOC-006 to v1.0 and create the Sprint 4 Git checkpoint/tag. No Sprint 5 implementation work should invalidate this baseline without an explicit design decision and, where required, controlled schema migration.

Appendix A. Core Error Catalogue

Code	HTTP	Retryable	Meaning
malformed_json	400	No	Malformed JSON request.
authentication_required	401	No	Authentication is required.
guest_session_invalid	401	No	Guest session invalid/expired/token-version mismatch.
insufficient_event_permission	403	No	Known principal lacks Event permission.
rsvp_window_closed	403	No	Guest RSVP window is closed.
resource_not_found	404	No	Resource absent or concealed.
resource_modified	412	Conditional	ETag/If-Match stale.
precondition_required	428	Conditional	Required If-Match omitted.
validation_failed	422	No	Field/schema/domain validation failed.
invitation_capacity_exceeded	422	No	Invitation capacity would be exceeded.
table_capacity_exceeded	422	No	Table capacity exceeded.
seat_already_occupied	409	Conditional	Seat is currently occupied.
guest_response_modified	409	Conditional	Guest response revision is stale.
seating_recommendation_stale	409	Conditional	Recommendation input changed.
seating_group_override_required	409	Conditional	Required grouping would be violated.
attendee_already_checked_in	409	No	Attendee is already effectively checked in.
checkin_already_reversed	409	No	Check-in already reversed.
campaign_already_queued	409	No	Campaign has already been frozen/queued.
message_not_retryable	409	No	Message is not eligible for retry.
import_not_ready	409	Conditional	Import has unresolved readiness conditions.
idempotency_key_conflict	409	No	Same key was reused for a different canonical request.
idempotency_request_in_progress	409	Yes	Matching idempotent request is still processing.

Code	HTTP	Retryable	Meaning
rate_limit_exceeded	429	Yes	Caller exceeded rate limit.
internal_error	500	Conditional	Unexpected server failure; use request_id for diagnostics.
temporarily_unavailable	503	Yes	Known temporary infrastructure/service issue.